

Alzheimer's Disease and Aging: Aims 1 and 2

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Aim 1 evaluate whether peripheral inflammatory markers are associated with a 1 year decline in memory and cortical thickness in aMCI. Hypothesis 1a tests whether baseline cytokine (IL-6, TNF- α) and chemokine (MCP-1, Eotaxin-1) levels predict decline. Hypothesis 1b tests whether changes in these markers over the 1 year period are associated with decline. For both hypotheses, the outcome includes AD-signature cortical thickness (continuous). Hypothesis 1a also focuses on memory consolidation (continuous); hypothesis 1b also focuses on five episodic memory outcomes (continuous; (1) 20' Delay, Total Correct, (2) Recognition, D-Prime, (3) 15' Delay, Total Correct, (4) Proportion Correct, (5) 1-Week Delay, Total Correct). All outcomes will be modeled as year 1 values and models will include baseline value adjustments, allowing coefficients to be interpreted as the effect on the 1 year change. For hypothesis 1a, the primary predictor is the baseline cytokine or chemokine value; for hypothesis 1b it is the 1 year change in that marker. Each inflammatory marker will be analyzed in separate models to avoid multicollinearity, with exploratory joint models. The general model form for aim 1 is: $Outcome_{Y1} = \beta_0 + \beta_1 Marker_{Y1 \text{ or } Y1-Y0} + \beta_2 Outcome_{Y0} + \beta_n Covariates + \epsilon$ with an alternative hypothesis that $\beta_1 < 0$.

Aim 2 examines whether amyloid deposition modifies the relationship between inflammation and clinical progression. The primary hypothesis tests whether baseline amyloid deposition interacts with baseline inflammatory markers to predict a one year decline in episodic memory and cortical thickness. The secondary hypothesis tests whether baseline amyloid deposition

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interacts with the 1 year change in inflammatory markers to predict the one year decline in the outcomes. Models follow a similar approach to aim 1, but include baseline amyloid and an interaction. The general model form for aim 2 is: $Outcome_{Y1} = \beta_0 + \beta_1 Marker_{Y1 \text{ or } Y1-Y0} + \beta_2 Amyloid_{Y0} + \beta_3 Marker_{Y1 \text{ or } Y1-Y0} * Amyloid_{Y0} + \beta_4 Outcome_{Y0} + \beta_n Covariates + \epsilon$ with an alternative hypothesis that $\beta_3 < 0$.

All models will adjust for age, sex, and baseline outcome *a priori*. Data-driven approaches (e.g. LASSO) will be used to identify additional covariates to include that are highly correlated among education, APOE, baseline CDR, GDS, BMI, hypercholesterolemia, blood pressure, NSAID use, immune-related health conditions, and cardiovascular risk. Interpretation will be cautious among correlated variables. For both aims, type I error will be controlled using Benjamini-Hochberg FDR, applied separately within each hypothesis across the different combinations of outcomes and inflammatory markers.

For power, each hypothesis will be approximated using a linear regression framework with a continuous predictor (inflammatory marker or amyloid*marker interaction). Because all longitudinal analyses model year 1 outcomes adjusted for baseline values, we can express the effect of interest as the partial R^2 attributable to the predictor of interest. This can be approximated as the squared correlation coefficient (with the effective sample size adjusted for baseline correlation with the year 1 outcome). Power analysis can then be conducted to determine the observable effect size at various power levels with the expected sample size (125 aMCI/50 HC after attrition). This approach will provide a conservative estimate of power, as additional covariates such as age and sex will function as precision variables in the final analysis. Separate power calculations will be performed for each hypothesis.

Appendix

Code Directory: <https://github.com/eweible/BIOS6624.git>

File name: P2/Code/P2_Data_Analysis_Plan.Rmd